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1 Likelihoods

What is a Likelihood?

$$y_i \stackrel{\text{iid}}{\sim} N(x_i^\top \beta, \sigma^2)$$

$$f(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i - x_i^\top \beta)^2}$$

That's just a height given a certain y_i . The likelihood is a product of function heights.

$$f(y_1 \cap y_2) = f(y_1)f(y_2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_1 - x_1^\top \beta)^2} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_2 - x_2^\top \beta)^2}$$

Likelihoods often mentioned as probabilities, and while they technically aren't, they basically are (see Measure Theory). For example, in Bayes':

$$Post(\beta|data) = \frac{Lik(data|\beta)P(\beta)}{\int \int Lik(data|\beta)P(\beta)d\beta}$$

where we are effectively treating likelihood as a probability.

$$L(\mathbf{y}) = \prod_{i=1}^n f(y_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i - x_i^\top \beta)^2} = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{n/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x_i^\top \beta)^2}$$

$$\sum_{i=1}^n (y_i - x_i^\top \beta)^2 = (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) \text{ This is then a dot-product of the two...}$$

$$\Rightarrow L(\mathbf{y}) = (2\pi)^{-n/2} (\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)}$$

2 MLE of β

$$\underset{nx1}{\mathbf{y}} = \underset{nxp}{\mathbf{X}} \underset{px1}{\beta} + \underset{nx1}{\epsilon}, \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \mathbf{y} \sim N(\mathbf{X}\beta, \sigma^2 I_n)$$

$L(\mathbf{y}|\beta, \sigma^2)$ we just switch the conditioning around for the likelihood of the parameters:

$$L(\beta, \sigma^2|\mathbf{y}) = (2\pi)^{-n/2} (\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)}$$

To find the MLE for β : $\frac{\partial}{\partial \beta} L(\beta, \sigma^2|\mathbf{y})$, set it equal to 0

$$l(\beta, \sigma^2|\mathbf{y}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)$$

When we take the derivative we will get p items because there are p elements in β .

$$\frac{\partial}{\partial \beta} l(\beta, \sigma^2|\mathbf{y}) = -\frac{1}{2\sigma^2} \frac{\partial}{\partial \beta} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)$$

$$(\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}^\top \mathbf{y} - \beta^\top \mathbf{X}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X} \beta + \beta^\top \mathbf{X}^\top \mathbf{X} \beta = \mathbf{y}^\top \mathbf{y} - 2\beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta$$

$$\frac{\partial}{\partial \beta} \mathbf{y}^\top \mathbf{y} - 2\beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \beta$$

$$\text{Set it equal to 0: } -2X^\top \underset{\sim}{\mathbf{y}} + 2X^\top X \underset{\sim}{\hat{\boldsymbol{\beta}}} = 0$$

$$\implies X^\top X \underset{\sim}{\hat{\boldsymbol{\beta}}} = X^\top \underset{\sim}{\mathbf{y}}$$

$$\implies (X^\top X)^{-1} X^\top X \underset{\sim}{\hat{\boldsymbol{\beta}}} = (X^\top X)^{-1} X^\top \underset{\sim}{\mathbf{y}}$$

$$\implies \underset{\sim}{\hat{\boldsymbol{\beta}}} = (X^\top X)^{-1} X^\top \underset{\sim}{\mathbf{y}}$$

$$\mathbb{E}(\underset{\sim}{\hat{\boldsymbol{\beta}}}) = \mathbb{E}[(X^\top X)^{-1} X^\top \underset{\sim}{\mathbf{y}}] = (X^\top X)^{-1} X^\top X \underset{\sim}{\boldsymbol{\beta}} = \underset{\sim}{\boldsymbol{\beta}}$$

$$V(A \underset{\sim}{\mathbf{y}}) = AV(\underset{\sim}{\mathbf{y}})A^\top$$

$$Var(\underset{\sim}{\hat{\boldsymbol{\beta}}}) = Var[(X^\top X)^{-1} X^\top \underset{\sim}{\mathbf{y}}]$$

$$\underset{pxn}{A} = (\underset{pxp}{X^\top X})^{-1} \underset{pxn}{X^\top}$$

$$\underset{nxp}{A}^\top = X(X^\top X)^{-1}$$

$$Var(\underset{\sim}{\hat{\boldsymbol{\beta}}}) = (X^\top X)^{-1} X^\top Var(\underset{\sim}{\mathbf{y}}) X (X^\top X)^{-1} = (X^\top X)^{-1} X^\top \sigma^2 I_n X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}$$

$$\underset{px1}{\underset{\sim}{\hat{\boldsymbol{\beta}}}} \sim N(\underset{px1}{\underset{\sim}{\boldsymbol{\beta}}}, \underset{pxp}{\sigma^2 (X^\top X)^{-1}})$$

3 Multi-variate Normal

$$L = (2\pi)^{-n/2} |\Sigma|^{-1/2} e^{(-\frac{1}{2}(y-x\beta)^\top \Sigma^{-1}(y-x\beta))}$$

$$\Sigma = \sigma^2 I_n$$

$$L = (2\pi)^{-n/2} (\sigma^2)^{-n/2} e^{(-\frac{1}{2\sigma^2}(y-x\beta)^\top (y-x\beta))}$$

$$l = -\frac{n}{2} \ln(2\pi) - \frac{1}{2} \ln(|\Sigma|) - \frac{1}{2} (y-x\beta)^\top \Sigma^{-1} (y-x\beta)$$

$$\frac{\partial}{\partial \beta} (y-x\beta)^\top \Sigma^{-1} (y-x\beta)$$